RONIT SAHU

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SUMMARY

B.Tech Robotics and AI student (2027) with project experience in Computer Vision, NLP, Agentic AI, RAG and LLM-based applications. Built a multi-mode YOLOv8 surveillance system with FastAPI backend, a real-time interview analysis system, and a Groq API pipeline for structured data extraction.

EDUCATION

Bangalore Institute of Technology, Bengaluru | Aug 2023 – Sep 2027 (Expected)
B.Tech in Robotics and Artificial Intelligence
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Agentic AI
12th (Science): 75% | **10th:** 90%

PROJECTS

SENTINEL – Multi-Mode AI Surveillance System | 2025 | *Python, YOLOv8, OpenCV, FastAPI*
- Built a surveillance analytics system with 4 modes: Airport & Defence, Market & Shop, Traffic, and Crowd & Event Control.
- Implemented YOLOv8 + OpenCV for person and vehicle tracking, crowd counting, congestion analytics and crowd-motion assessment, served via FastAPI.
- Automates threat monitoring and customer-flow analysis, reducing dependence on manual CCTV monitoring.

Comprehensive Interview Analysis System | 2026 | *Python, OpenCV, NLP*
- Built a real-time system that evaluates 3 signals per candidate: sentiment, verbal communication quality, and facial cues.
- Combined OpenCV facial-cue detection with NLP models to score candidates during live interviews, giving recruiters consistent, objective assessments.

LLM-Based Customer Ticket Information Extraction | *Groq API, Pydantic, Python*
- Built an LLM workflow that extracts customer information from unstructured support tickets into structured JSON.
- Defined a Pydantic schema and validated every parsed LLM output before downstream use, removing manual data entry.

SKILLS

Languages: Python, Java, C, C++, SQL
AI/ML Concepts: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Agentic AI, RAG, ANN, CNN, RNN, LSTM, GRU
Frameworks & Tools: TensorFlow, Keras, Scikit-Learn, OpenCV, YOLOv8, LangChain, LangGraph, CrewAI, ChromaDB, FastAPI, Groq API, Pydantic, MySQL

CERTIFICATIONS

Machine Learning Foundations & Implementation